

Rahul Tammalla

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Experience

Research assistant, Clarkson University, – Potsdam, United States. Jan 2025 – Present

- Analyzed 30 million healthcare records on **Nemsis** dataset to optimize medical response times using data analysis and machine learning techniques.
- Conducted data preprocessing (cleaning, transforming, handling missing values) to ensure data quality and integrity across datasets.
- Performed feature engineering, statistical analysis, and hypothesis testing to identify key predictive factors affecting on-scene response time less than 10 mins.
- Using **PostgreSQL** for database management, query optimization, and developing interactive Tableau visualizations to present actionable insights and improve resource allocation.

Associate Engineer, Cognizant Technology Solution, – Hyderabad, India. Oct 2021 – Aug 2024

- Optimized and engineered a high performance cloud-based Appian application for **Gilead Life Sciences**, a leading pharmaceutical company, processing 1M+ transactions to enhance reliability and user experience.
- Designed interactive dashboards for real-time insights, improving decision-making and visibility into key metrics.
- Improved Database and **Store procedure** performance, increasing execution speed and response time by **30%**, and ensured data integrity through daily system file analysis.
- Performed a daily in-depth data analysis on **500+** daily system records (transactions, inventory, on-hand, delivery) to ensure accuracy and seamless operations.

Education

Clarkson University, Masters in Applied Data Science (GPA : 3.889) Aug 2024 – Dec 2025

- **Coursework:** Probability Statistics, Information visualization, Database modelling, Machine Learning, Data Mining, Data warehousing.

Amrita Vishwa Vidyapeetham, Bachelors in Computer Science Engineering May 2017 – June 2021

Projects

Personalized Course Recommendation Database System 2024

- Developed a dynamic recommendation system with SQL triggers and stored procedures for automated course suggestions, and optimized database structures (views) for exam schedules, instructor workloads, and student performance, improving academic planning and resource management.

US Accident Insights 2024

- Developed the "US Accident Insights" Shiny app to analyze 40K+ records and visualize traffic accident data, using R libraries (dplyr, lubridate, ggplot2) for data preprocessing and visualization, and integrating interactive Shiny and Leaflet features for real-time actionable insights to improve road safety strategies.

Superstore Sales Report 2024

- Preprocessed data by handling missing values and ensuring consistency, then created a stacked bar graph to analyze profits and losses by region and product category, providing actionable insights.

Skills

Programming & Development: Python, SQL, R, Java, MATLAB, Agile, Scrum, OOP, HTML & CSS, MS tools.

Libraries & Frameworks: NumPy, Pandas, SciPy, scikit-learn, PySpark, TensorFlow, Keras, OpenCV, Matplotlib, Seaborn, AWS, Azure.

Data Engineering & Visualization: ETL, CI/CD, DevOps, R Studio, Tableau, Power BI.

Databases & Cloud: PostgreSQL, MongoDB, Snowflake, Hadoop, Hive, Azure Databricks.

Machine Learning : Regression, Classification, Clustering, Random Forests, XGBoost, SVM, Statistical Modeling, Ensemble Learning, CNN, Neural Networks, Time series.